

Project AegisTrader: Growing Trading Personalities

AegisTrader is not a backtesting platform with AI features. It is a platform for **growing, experimenting on, and coaching trading personalities**. The backtest engine provides credibility. The soul provides the experience. Users choose a trading philosophy, give it life experience through historical markets, watch it develop beliefs and doctrine, fork it into alternate timelines, and coach it toward better judgment.

GOVERNING PRINCIPLE

"Soul powered by backtest."

Every soul claim traces back to deterministic, reproducible evidence. The engine provides reproducibility and trust. The soul provides meaning and differentiation. Neither works without the other.

THE PRODUCT LOOP

Choose. Grow. Coach.

Three moves that define the AegisTrader experience, forming a continuous cycle of personality development.



Choose

Pick a trading philosophy archetype and give it life

MOVE 1: SOUL



Grow

Fork beliefs, run experiments, explore alternate timelines

MOVE 2: FORK





Coach

Shape development through review, guidance, and intervention

MOVE 3: COACH

MOVE 1

Soul as the Product

The trading soul — not the backtest engine — is the primary object of user attention. "**Grow trading personalities that learn from experience**" is the pitch. This creates a product category that does not exist.



Soul-First Onboarding Flow

MINUTE 0–2

Archetype Picker

Choose a trading philosophy: **Mean Reversion**, **Trend Following**, **Event-Driven**, or **Defensive Value**. Each archetype is a pre-built strategy.md + seed soul.md. The user is choosing a worldview, not configuring software.

MINUTE 2–5

Formative Experience

System auto-selects a dramatic historical period and runs a compressed 3–6 month backtest. The UI emphasizes soul formation: *"Your agent just experienced its first 30% drawdown. It's forming its first scar tissue."*

MINUTE 5–10

Soul Diff — The Hero Screen

Before/after comparison showing how the agent changed. Every belief is evidence-linked: *"This agent now reduces size after consecutive losses — that came from losing 12% in week 3."*

BRIDGE

Fork Prompt

"Want to keep training? Or fork this soul and see what happens if it had a different experience?" — bridges directly to counterfactual forking.

+ The Reframe: Before vs. After

BEFORE (PRD)	AFTER (SOUL-FIRST)
"Run backtests, see results"	"Grow a trading personality, watch it evolve"
Backtest results are the output	Soul evolution is the output
Branch DAG is debugging	Branch DAG is biography
PnL charts are the hero	Soul diff is the hero
User is an operator	User is a curator / coach

+ Critical Constraints

- **Seed archetype quality is non-negotiable.** Each archetype's strategy.md must be a genuinely viable trading approach, not a demo. A bad first backtest kills the onboarding.
- **Evidence-linking prevents "gimmick" perception.** Without concrete trade-to-belief links, the soul-first framing risks feeling like marketing rather than substance.

MOVE 2

Counterfactual Soul Forking

Fork an agent's **beliefs** — not just its trades — to run "what if" experiments on trading psychology itself. The branch DAG becomes a psychology lab. The soul becomes a research subject the user experiments on.

MVP

Trauma Test

Question: "What if my agent started in 2019 vs 2022?"

Clone the same archetype. Run one through a bull market era, one through a bear market era. Compare the resulting souls side by side.

The bull-market agent becomes aggressive and naive about drawdowns. The bear-forged agent becomes cautious and misses rallies.

Neither is "right" — the divergence teaches how experience shapes doctrine.

Implementation: Same strategy, different time ranges, compare resulting souls. Builds on existing branch DAG.

V2

Selective Amnesia

Question: "What if my agent forgot its worst month?"

Remove a specific period from a mature soul's experience. Re-derive the soul from branch history minus the excluded period.

Tests whether specific scars are **wisdom or damage**. "*Which scars are useful and which are just damage?*"

Implementation: Non-trivial. Beliefs are interconnected. Requires re-derivation from parameterized experience set.

+ Architecture Impact

- **New entity: "soul fork"** — distinct from branch (branch = alternate trades within a run, soul fork = alternate life experiences across runs)
- Soul comparison infrastructure with divergence ranking and prioritization
- Experience-set parameterization for soul derivation
- Multi-run stability checking for counterfactual claims

+ Critical Constraints

- **LLM non-determinism could undermine counterfactual claims.** The same agent run through the same period twice may produce different soul reflections due to LLM variance. Mitigation: run each condition multiple times and surface only stable, reproducible differences.
- **Comparison UX must surface top 3–5 divergences,** not dump everything. The screen needs intelligent summarization.

MOVE 3

User as Coach

Replace the binary approve/reject model with a **coaching paradigm**.

The user is a coach developing a trading personality, not an admin approving file changes.

MVP

"Let it Learn"

PASSIVE MODE

Post-game film review.
User reviews soul diffs
after runs, accepts or
flags specific beliefs.

Reframe: *"Reviewing a
lesson plan,"* not
"approving a patch."

V1.5

"Guided Reflection"

ACTIVE MODE

User provides natural
language coaching
before soul update
commits: *"You're over-
weighting the losses in
week 3."*

The user shapes the
reflection, not the
conclusion. This is
where most users find
the most value.

MVP

"Soul Surgery"

DIRECT MODE

Power-user direct
editing of soul.md /
soul.json. Manual edits
source-tagged as
`coach_override`.

The agent distinguishes
coached beliefs from
self-learned ones.

Soul Studio UI

V2

LEFT: SOUL TIMELINE

Version history color-coded by source. Self-learned = blue, coached = green, counterfactual = purple. Aggressive filtering from day one.

CENTER: BELIEF CARDS

Current soul as interactive cards. Each shows belief, confidence, evidence count, and a "Challenge" button for coaching intervention.

RIGHT: EVIDENCE DRAWER

Supporting trades, branches, runs, and PnL impact for any selected belief. The receipts behind every claim.

Retention Hook: "Am I a Good Coach?"

Track coached beliefs vs. organic beliefs over time. Surface coaching effectiveness: *"Your coaching improved returns by 12% over what the agent would have learned on its own."*

Per-belief attribution shows which user intuitions are actually correct. Turns engagement from passive observation into a **trackable skill**.

+ Critical Constraints

- **Full Soul Studio is too heavy for MVP.** Start with simplified belief cards and soul diff review, add three-panel layout in v2.
- **Source tagging in soul.json is non-negotiable from day one.** Without `source` fields from the start, retrofitting provenance is painful.
- **The agent must push back on coaching.** If it rubber-stamps every override, users lose trust. *"Evidence from runs #23-#31 supports my current belief. Are you sure?"*

SIGNATURE UX

Soul Comparison — The Convergence Screen

The side-by-side soul comparison screen is where **all three moves converge into a single experience**. It is the screen where AegisTrader's identity crystallizes for the user.

Soul Comparison: Bull vs. Bear Experience

Same archetype (Trend Following). Different formative eras. Top 3 divergences shown.

BULL-MARKET AGENT

2019–2021

VS

BEAR-MARKET AGENT

2022–2023

DIVERGENCE

POSITION SIZING

Aggressive scaling — "momentum rewards conviction." Increases size during winning streaks.

DRAWDOWN RESPONSE

Hold through dips — "drawdowns are buying opportunities." Max 15% drawdown experienced.

RISK TOLERANCE

High — never learned fear. Lost 8% max in any single week.

POSITION SIZING

Conservative scaling — "protect capital first." Reduces size after consecutive losses.

DRAWDOWN RESPONSE

Cut losses quickly — "drawdowns compound." Max 35% drawdown experienced.

RISK TOLERANCE

Low — "scar tissue from losing 12% in week 3." Misses 40% of rallies.

All three moves in one interaction:

[View Evidence](#)

[Fork Again](#)

[Coach This Soul](#)

On this screen, the user simultaneously:

- **Sees two souls shaped by different experiences** (counterfactual forking)

- **Reads their divergent beliefs with evidence links** (soul-as-product)
- **Can challenge or coach either one** (coaching UX)

Design priority: This screen should receive outsized design investment. It is not a comparison utility — it is the product's defining interaction.

Key Design Decisions

1. Evidence Chain as Shared Infrastructure

All three moves depend on **structured evidence provenance for every belief in soul.json**. This is the architectural foundation.

MOVE	WHAT IT NEEDS FROM EVIDENCE CHAIN
SOUL DIFFS	Verifiable links from each belief to the trades/runs that produced it
COUNTERFACTUAL	Distinguishing genuine experience-driven divergence from LLM noise
COACHING	Measuring whether coached beliefs outperform self-learned ones

+ required_evidence_refs[] Schema

- run_id / branch_id — which run or branch produced this evidence
- trade_ids[] — which specific trades support the belief
- branch_comparisons[] — which branch-vs-branch comparisons tested it
- confidence — confidence level based on sample size
- source_type — self_learned | coach_override | coach_guided | counterfactual_derived
- timestamp — when this evidence was incorporated

MVP requirement: The evidence chain schema must ship in Phase 1. Retrofitting structured provenance onto flat evidence lists is architecturally painful.

2. Deterministic Boundary Stays Sharp

The two-layer architecture is preserved. The soul-first framing makes the deterministic engine *more* important, not less — it is the credibility layer.

The non-deterministic layer may only produce proposals, annotations, and hypotheses. The deterministic layer alone produces accepted decisions, fills, and branch outcomes.

3. Source Tagging From Day One

Every belief in soul.json carries a `source` field from MVP launch. Non-negotiable.

Enables: agent pushback on coaching, coaching effectiveness measurement, counterfactual provenance, and soul integrity over time.

Competitive Positioning

AegisTrader targets a white space where no existing platform operates.
Algorithmic trading market: **\$24B (2025) projected to \$44.5B (2030)**,
13.2% CAGR.

Moat: Four Things No Competitor Offers Together

PLATFORM	DETERMINISTIC SIMULATION	EVOLVING SOULS	BRANCH DAG	COACHING UX
AegisTrader	Yes	Yes	Yes	Yes
QuantConnect	Yes	No	No	No
Composer	Partial	No	No	No
NautilusTrader	Yes	No (by design)	No	No
Numerai	No	No	No	No
AI Agent Frameworks	No	Emerging	LangGraph only	No

+ Key Market Gaps AegisTrader Fills

- **No platform combines deterministic backtesting with AI agent exploration.** QuantConnect has backtesting but no agents. Agent frameworks have agents but no simulation core.
- **No platform offers branch DAG exploration for trading strategies.** Git-like history for trading decisions is genuinely novel.
- **No platform has "trading soul" evolution.** Agents developing durable, versioned, evidence-linked doctrine over time is unique.
- **No platform bridges natural language strategy definition with deterministic execution.**
- **Multi-agent competition within a personal research environment** is not offered by any existing product.

+ Competitive Risks

- QuantConnect could add AI features (backtesting engine + community already exist)
- LangChain/LangGraph ecosystem could release a finance-specific template
- Alpaca or other brokers could build a first-party AI research layer
- General AI coding agents could reduce perceived need for a specialized platform

Boundaries

What AegisTrader is **not**.

Not a Brokerage

AegisTrader does not execute real-money trades in initial versions. Paper trading and research tooling only.

Not Financial Advice

The platform must clearly label itself as paper trading and research tooling. No investment recommendations.

Not an Opaque AI Black Box

Every soul belief links to evidence.
Every decision is inspectable. The soul is interpretation, not truth.

The Soul Is Not the Source of Truth

The **deterministic engine** is. The soul is a learned interpretation of truth — derived, versioned, diffable, and replay-linked.

Phasing Roadmap

The existing PRD phases remain valid. The soul-first framing reshapes what ships in each phase and adds a v1.5 milestone.

PHASE 1 — MVP

Single-Agent Soul Evolution

Archetype picker with 4 trading philosophies. Compressed formative experience backtest. Soul diff as hero screen with evidence links. Trauma Test counterfactual. Side-by-side soul comparison with divergence ranking. Passive coaching + Soul Surgery. Source tagging and structured evidence_refs in soul.json.

PHASE 1.5

NEW

Guided Coaching

Guided Reflection mode (natural language coaching). Challenge button on belief cards. Enhanced Soul Studio layout.

PHASE 2

Multi-Agent Competition

As currently defined in PRD, plus coaching effectiveness tracking across agents and per-agent soul comparison.

PHASE 3

Live Paper Trading

As currently defined, plus live soul updates with coaching review before commit and coaching effectiveness dashboard.

PHASE 4

Governance & Advanced Features

Selective Amnesia counterfactual. Full coaching effectiveness analytics. Evaluate Adversarial Soul Dynamics based on user demand.

+ New Elements to Add to PRD

ELEMENT	TARGET
Archetype templates (strategy.md + seed soul.md)	MVP
Soul fork entity in data model	MVP
Structured evidence_refs[] in soul.json	MVP
Source field in soul.json schema	MVP
Side-by-side soul comparison screen	MVP
CoachingInput, BeliefProvenance entities	MVP
Guided Reflection mode	V1.5
Full Soul Studio three-panel UI	V2
Selective Amnesia counterfactual	V2
Coaching effectiveness dashboard	V2

+ Deferred Ideas

IDEA	REASON DEFERRED	REVISIT
Adversarial Soul Dynamics	Premature complexity; multi-agent basics need to work first	Phase 3/4
Cross-Training Counterfactual	Unclear user value; academic curiosity	User demand
Broad Belief Graduation	Risks eroding the deterministic boundary	If narrow version works

Open Questions

Key tensions requiring empirical resolution.

1. LLM Non-Determinism in Counterfactuals

How many runs are needed to establish stable soul divergences? What's the threshold for surfacing a difference as "real" vs. noise? Needs empirical testing during Phase 1.

2. Guided Reflection Quality

Can LLMs reliably integrate vague natural language coaching ("*you're too cautious*") into coherent, structured soul updates? Determines whether v1.5 ships as designed.

3. Soul Timeline Noise Management

After 20+ runs, the soul timeline will be dense. What summarization and grouping strategies keep it usable? Must ship with filtering from day one.

4. Archetype Design

Who designs the seed strategy.md + soul.md for each archetype? These must be genuinely viable strategies — they are the product's first impression.

5. Coaching Effectiveness Validity

How much run history is needed before coaching effectiveness metrics are statistically meaningful? Early results may be noisy.

Participants: Free Thinker, Grounder, Explorer, Writer

Sources: docs/Concept_1.md, docs/PRD.md, 2 competitive research reports, 3 briefs, 5 snapshots